An experimentally validated prototype study of sim-to-real deployment for conveyor-based coffee bean sorting under pneumatic actuation constraints

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Abstract: Deep learning (DL) technology has assumed a prominent role in the defect inspection of agricultural products. However, the deployment of these models in dynamic environments frequently exposes a critical sim-to-real gap. This research presents an experimentally validated prototype study of a green coffee bean sorting apparatus, conducted via constraint-driven analysis, which integrates the You Only Look Once (YOLOv12) computer vision architecture with a pneumatic actuation mechanism. Evaluations conducted on static image datasets and physical conveyor belt tests indicated that, although the baseline model achieved elevated accuracy levels on static images, dynamic operation induced temporal prediction instability. This instability was characterized by class oscillation and a measured increase in false rejections. This discrepancy was mitigated at the level of observed system behavior through structural hyperparameter tuning, specifically via resolution scaling and targeted active data augmentation. The optimized prototype increased vision-sensor-only classification accuracy on the physical conveyor from 83.07% to 89.60%, while reducing false rejections within the vision system evaluation tier by 46%. However, these figures represent evaluations strictly at the vision-system level and must not be extrapolated to represent the pneumatic-integrated performance of the entire apparatus. The findings indicate that in a sorting prototype equipped with pneumatic actuation, model performance must be evaluated in conjunction with hardware latency constraints and mechanical variance to mitigate actuation errors. The obtained results present a conceptual framework for analyzing the correlation between computational vision capabilities and physical constraints within agricultural sorting applications.

Keywords: Coffee sorting; YOLOv12; Sim-to-real gap; Pneumatic actuation; Machine vision; Smart agriculture.

1. Introduction

Coffee is considered one of the most globally significant economic crops with substantial trading value. The quality of green coffee beans is an indicator that directly impacts the flavor profile and market value of the final product. However, defective beans are invariably present during the harvesting and processing stages. The presence of even a minute quantity of defective beans can significantly degrade quality and result in the coffee being downgraded. Consequently, the coffee bean sorting process is a critical step in the coffee production industry. Currently, the majority of the industry relies on a multi-stage sorting process, initiating with a gravity separator and an optical color sorter for preliminary screening. Nevertheless, these processes exhibit limitations in detecting complex defects or minuscule blemishes with colors akin to normal beans. Ultimately, coffee roasters and producers must still rely on manual sorting as a final step; a method that incurs high costs, consumes considerable time, and is susceptible to human error caused by fatigue and visual assessment inconsistencies.

To overcome these limitations, computer vision and deep learning technologies have been widely applied to develop automated sorting prototypes. Yet, the majority of current research frequently halts at laboratory simulations or static dataset evaluation, engendering a sim-to-real gap. When models demonstrating measured accuracy in computational environments are deployed on physical moving conveyor belts, they frequently encounter dynamic disturbances such as motion blur, illumination variations, and class flickering. These phenomena cause the processing system to transmit erroneous actuation signals to the rejection hardware. Therefore, this research focuses on systematically studying and mitigating the sim-to-real gap by presenting an experimental analysis of a sorting prototype. This apparatus integrates the YOLOv12 neural network architecture with a pneumatic actuation mechanism, utilizing synchronized vision processing in conjunction with a microcontroller to investigate the constraint-driven behavior of the system within a simulated dynamic environment.

1.1 Research Contributions

This research presents the following engineering contributions.

1. Experimentally Validated Prototype Study: Presents an evaluation of the integration between the YOLOv12 architecture and pneumatic hardware under the physical constraints of a conveyor belt.
2. Constraint-driven Sim-to-Real Analysis: Identifies the root causes of operational discrepancies, providing empirical evidence that these deviations stem not solely from computational vision but are correlated with the mechanical latency of the hardware.
3. Dynamic-Aware Optimization Strategy: Proposes a hybrid training strategy observed to increase tolerance to visual noise, which correlates with a reduction in false rejections at the observed phenomenon level.
4. Engineering Bottleneck Identification: Provides insights from integrated testing, isolating algorithmic limitations from the physical constraints of sensors and pneumatic cylinders to establish baseline data for future scaling.

1.2 Related works

Recent advancements in automated green coffee bean inspection indicate a transition towards utilizing deep learning architectures to improve defect detection and classification. Numerous studies have applied YOLO and CNN architectures. For instance, Chou et al. [2] augmented accuracy using GAN-based data augmentation techniques, while Silva et al. [4] and Thai et al. [5] demonstrated the potential of the YOLOv8 architecture in detecting coffee bean defects with measured precision. However, these systems are predominantly restricted to testing within controlled laboratory environments on static datasets and lack integration with mechanical actuators for physical sorting processes.

Beyond defect detection, deep learning applications encompass various dimensions of coffee bean quality assessment. Recent research has indicated the potential of deep net models in quality grading [8], [10], single-origin classification [9], and roasting level evaluation **via the** MobileNet architecture to support mobile device processing [7]. Yet, the majority of these evaluations are conducted under the constraints of static datasets, without addressing mechanical challenges and dynamic vision challenges.

While Chang and Huang [1] and Huang et al. [6] presented classifications with accuracy exceeding 94% through architectural modifications and the use of real-time IP cameras on a conveyor belt, practical deployment still encountered constraints. These included single-sided bean analysis and the absence of an automated physical rejection mechanism for good or defective beans on the production line. To elevate accuracy, Chen et al. [3] proposed a more complex approach utilizing multimodal NIR-Hyperspectral imaging combined with 2D-3D CNNs, yielding an accuracy

of 97.5–98%. Nevertheless, the cost and equipment complexity remain primary barriers to widespread commercial deployment compared to standard RGB systems.

Despite continuous efforts to develop automated prototypes, the realization of a cost-constrained, integrated apparatus remains a notable gap in current research. To address this gap, the novelty of this research is an experimentally validated prototype study that fine-tunes YOLOv12 hyperparameters to analyze its integration with hardware, focusing on resolving the sim-to-real gap between algorithmic modeling and physical execution. This research presents processing on a computer system temporally synchronized with a microcontroller to command a pneumatic actuation unit, facilitating mechanical constraint analysis under a simulated dynamic environment.

2. Proposed System and Methodology

The development of the automated coffee bean sorting prototype in this research relies on the cross-disciplinary integration of computer vision engineering and electro-pneumatic control systems. The methodological framework was designed to address real-world constraints. The architectural details of the system are presented as follows.

2.1. Hardware Architecture and Sorting Mechanism

The coffee bean sorting prototype was designed with spatial dimensions of 250 × 500 × 150 mm (width × length × height), as illustrated in Figure 1. The hardware architecture comprises three primary operational modules: the conveyor belt system, the vision detection system, and the pneumatic rejection mechanism.

Regarding the conveyor system, a white conveyor belt surface was specified to establish high color contrast between the target objects and the background. This serves as a physical control variable that mitigates the complexity of feature extraction for the model. The drive system utilizes an induction motor to maintain a constant linear velocity. Furthermore, the inspection zone was illuminated under controlled lighting conditions using a circular LED ring light to minimize illumination fluctuations and cast shadows during image acquisition.

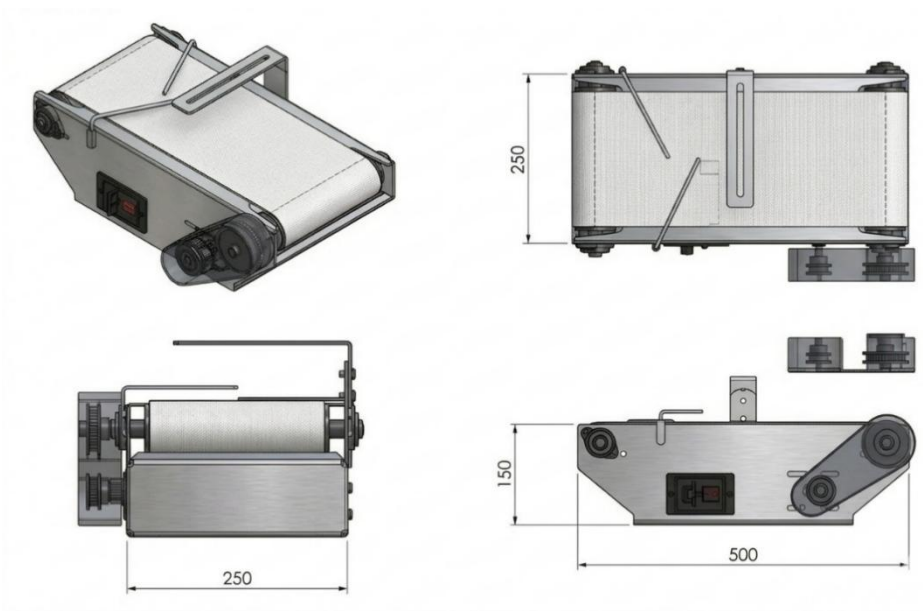


Figure 1. Structural design of the conveyor-based coffee bean sorting prototype.

Regarding the control circuit architecture illustrated in Figure 2 and Figure 3, an Arduino Uno R3 microcontroller functions as the central unit for mechanical actuation, controlling the 12-24V solenoid valves via a 5V relay module. The specified circuit was designed to incorporate electrical isolation utilizing PC817 optocouplers in conjunction with flyback diodes. This configuration is intended to maintain signal stability and prevent electrical damage to the processing unit. For the regulation of the conveyor belt motor, the apparatus utilizes an ESP8266 module operating in tandem with a relay.

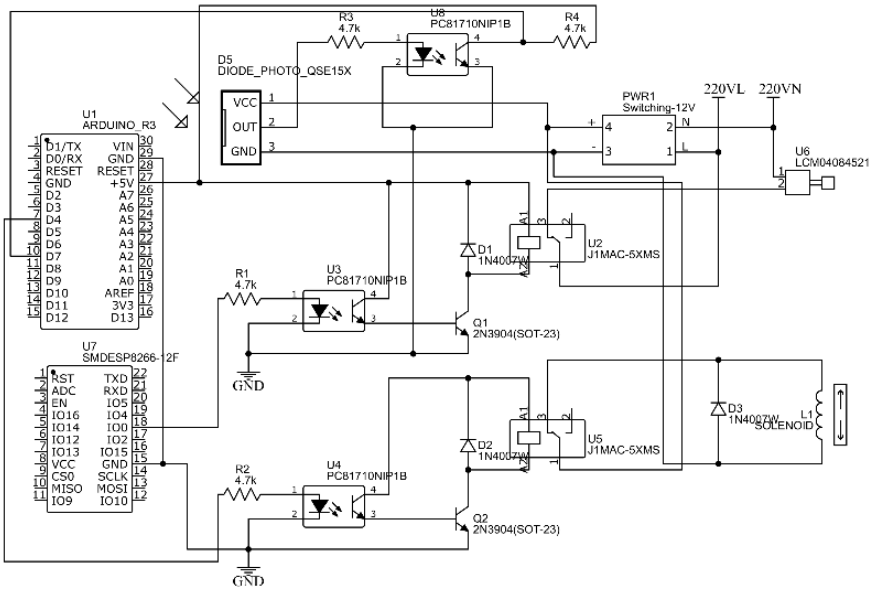


Figure 2. Schematic diagram of the control circuit for the coffee bean sorting machine.

The physical ejection mechanism operates via a pneumatic system comprising an air compressor, a solenoid valve, and a pneumatic cylinder. Upon the classification of a defective bean by the processing unit, the Arduino microcontroller transmits an electrica l actuation signal to the solenoid valve. This signal regulates the flow of compressed air into the cylinder, inducing the extension of the cylinder ro d to physically displace the targeted bean from the conveyor belt. The integrated hardware operational sequence is depicted in **Figure 3**.

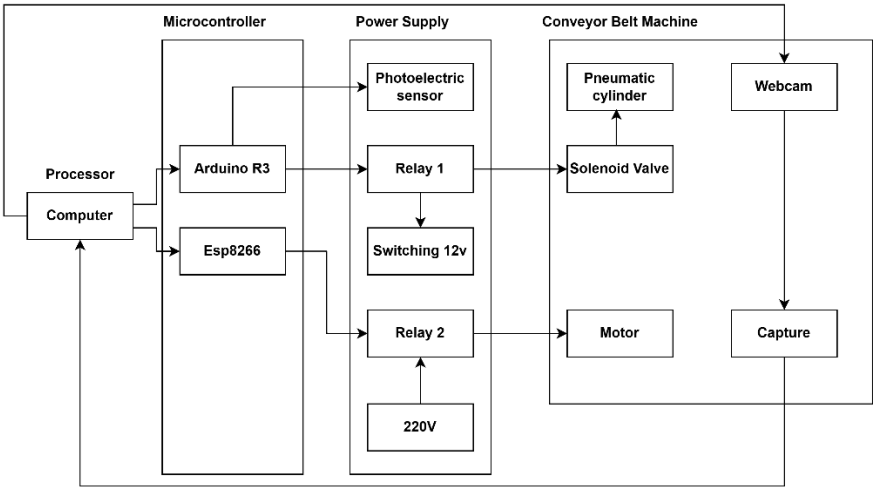


Figure 3. Block diagram of the automated coffee bean sorting system.

The operational sequence and hardware-software integration architecture are summarized in the schematic presented in Figure 4. The process initiates with the introduction of coffee beans onto the conveyor belt, maintaining spatial separation to minimize physical occlusion. As the objects transition into the inspection zone, the vision camera acquires images and transmits the streaming data to the processing unit for evaluation utilizing the YOLOv12 architecture.

During the actuation synchronization phase, a photoelectric sensor (E3Z-D62), positioned preceding the ejection point, detects the passage of the beans and transmits a hardware interrupt signal to the Arduino microcontroller. Subsequently, the system polls the command queue generated by the vision system. If the transmitted data designates the target within the "Defective" class, the microcontroller calculates a timing delay correlated with the linear velocity of the conveyor belt. Following this delay, the system commands the solenoid valve to channel pressurized air into the pneumatic cylinder, thereby physically displacing the target bean from the belt. Conversely, if the object is classified as "High-quality" (Good), the microcontroller bypasses the mechanical actuation sequence, permitting the product to proceed to the terminal collection point under standard operating conditions.

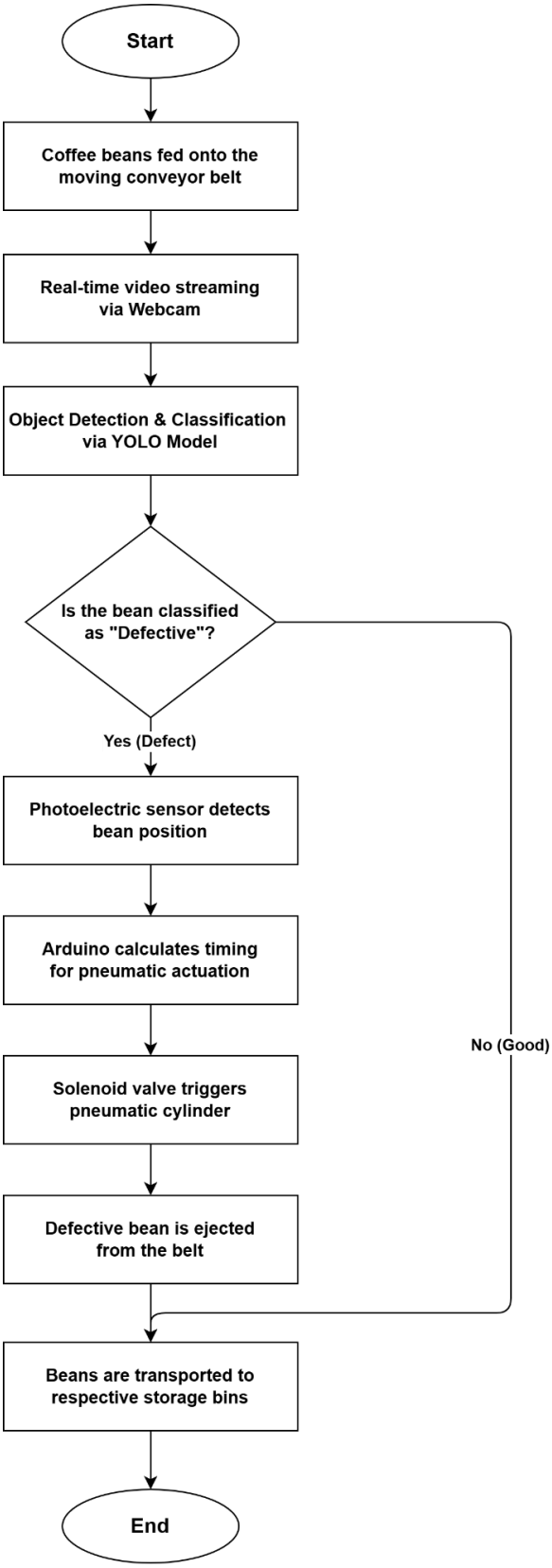
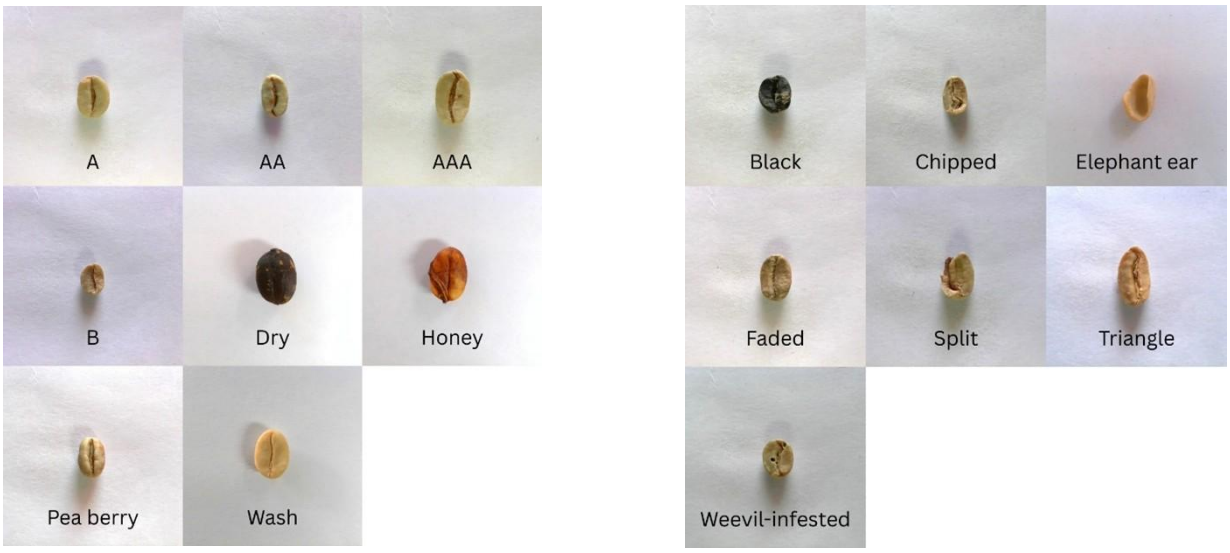


Figure 4. Operational flowchart of the hardware and software integration.

Figure 5. Physical setup of the conveyor belt and pneumatic sorting equipment.

2.2 Dataset Preparation

Multiple dataset versions were constructed to support both the baseline evaluations and the hyperparameter-tuning experimental phases. The coffee bean categories utilized as the dataset in this research were divided into 15 distinct classes: A, AA, AAA, B, Black, Chipped, Dry, Elephant ear, Faded, Honey, Peaberry, Split, Triangle, and Wash. The preparation of the training datasets was executed via the Roboflow platform, which served as the management infrastructure for processing the image data acquired by a consumer-grade RGB camera. Sample images of the categorized beans are depicted in **Figure 6**



(a) **(b)**

Figure 6. Examples of the 15-class green coffee beans dataset: (a) Good beans (A, AA, AAA, B, Dry, Honey, Pea berry, and Wash); and (b) Defective beans (Black, Chipped, Elephant ear, Faded, Split, Triangle, and Weevil-infested).

Images of the 15 coffee bean classes, encompassing both acceptable and defective categories, were partitioned into two primary dataset versions to facilitate comparative performance evaluation, as detailed in Table 1. The Baseline Dataset comprised 4,438 original images, establishing the standard benchmark for subsequent evaluations. Following the observation of misclassifications during preliminary mechanical testing, a Hyperparameter-Tune Dataset was developed. This involved the targeted acquisition of supplementary images specifically for classes

exhibiting elevated error rates, namely classes A, AA, AAA, Chipped, and Pea berry. These supplementary images were acquired under actual operational illumination conditions on the moving conveyor belt from multiple spatial perspectives. This targeted data collection expanded the total volume of original images to 5,638, explicitly intended to increase data variance within the specific classes subjected to model confusion

Table 1. Summary of dataset versions and image distributions.

Dataset Version	Number of Classes	Total Images	Notes
Baseline 15-Class	15	4,438	Original dataset
Hyperparameter-Tune 15-Class	15	5,638	Targeted data augmentation for problematic classes

The dataset construction was divided into two primary phases to evaluate the impact of data augmentation and hyperparameter t uning, as summarized in **Table 2** The Baseline dataset consisted of 4,438 original images, split into training (3,552), validation (443), and testing (443) sets, maintaining an approximate ratio of 80:10:10 without any external augmentation. To enhance model robustness and prevent overfitting under real-time dynamic conveyor conditions, the Hyperparameter-Tune dataset was generated using Roboflow.

Various data augmentation techniques including Auto-Orient, 90 Rotate (Clockwise, Counter-Clockwise, Upside Down), Hue (Between -20% and +20%), Brightness (Between 0% and +20%), and Blur (Up to 1.5px) were applied **strictly to the training set only**. This strategy prevents data leakage while exponentially expanding the total dataset to 26,332 images with an approximate ratio of 91:6:3.

Table 2. Dataset configurations and data augmentation strategies.

Dataset Version	Train Images	Valid Images	Test Images	Augmentation
Baseline 15-Class	3,552	443	443	None
Hyperparameter-Tune 15-Class	23,940	1,594	798	Roboflow Augmentation Applied

To delineate the specific composition of the experimental dataset, Table 3 summarizes the respective class categories and their corresponding image frequencies. This detailed breakdown is provided to elucidate the class balance and data distribution charac teristics, which constitute critical parameters governing the efficacy of the model training process.

Table 3. Class distribution in the dataset before and after hyperparameter tuning

Dataset Version	Number of Classes (Before)	Number of Classes (After)
A	305	505
AA	300	500
AAA	301	501
B	300	500
Black	301	301
Chipped	304	504
Dry	220	220
Elephant ear	300	300
Faded	301	301
Honey	301	301
Pea berry	304	504
Split	300	300
Triangle	300	300
Wash	300	300
Weevil-infested	301	301

2.3 AI Model Architecture, Training Strategy, and Confirmation Logic

In this research, three object detection model architectures were designed and experimentally evaluated: YOLOv8s, YOLO11s, and YOLOv12s. The utilization of multiple versions functions as a comparative model analysis to identify the optimal architecture for the automated coffee bean sorting application. The objective is to determine which architecture yields the highest measured accuracy and temporal prediction stability under the constraints of the prepared datasets.

2.3.1 Training Setup and Hyperparameters

To govern the training process, specific configurations and hyperparameters were established. The training phase was executed utilizing the Ultralytics YOLO framework within a GPU-accelerated environment (CUDA). The defined parameters encompass the total training epochs, batch size, initial learning rate, learning rate scheduler, data augmentation strategies, and the selected optimization algorithm. Detailed hyperparameter configurations are summarized in the subsequent **Table 4**

Table 4. Hyperparameter configurations for model training.

Model	Epochs	Image Size	Optimizer	LR (Initial)	Batch Size	Ultralytics Augmentation
Baseline	100	640	AdamW	0.001	8	HSV, Flip, Mosaic, erasing, translate, scale, shear
Hyperparameter-Tune	100	704	AdamW	0.0005	8	HSV, Flip, Mosaic, Mix up, degrees, copy paste, translate, scale

A primary rationale for increasing the input image resolution from 640 to 704 pixels in the hyperparameter-tuned model is to augment the network's spatial feature extraction capabilities for minute surface anomalies, specifically micro-cracks and weevil-induced perforations. Furthermore, the application of targeted data augmentation techniques, notably MixUp and rotational variance, was strategically implemented

to improve the model's robustness against dynamic motion blur. This approach was designed to mitigate the inter-class confusion prevalent in the empirical results of the baseline model.

Table 5. Training Environment

Component	Specification
CPU	Intel Xeon Processor @ 2.20GHz (Cloud Compute)
System RAM	167GB (High-RAM Instance)
GPU	NVIDIA A100 Tensor Core GPU (80GB VRAM)
Storage	SSD 235GB
Operating System	Ubuntu 22.04 LTS (Google Colab Environment)
IDE	Google Colaboratory

2.3.2 Confirmation Logic

To mitigate single-frame misclassification errors, the system implemented a Majority Voting logic, stipulating the continuous detection of a target object across five consecutive frames prior to transmitting an actuation signal to the solenoid valve, as illustrated in the class confirmation sequence in **Figures 7–9.** Object detection and tracking were executed utilizing the Ultralytics YOLO library via the model.track function. This function performs simultaneous detection and tracking to classify the object and monitor the spatial displacement of the coffee beans on the conveyor belt.

The system incorporated the ByteTrack algorithm in conjunction with the detection model to facilitate continuous multi-frame tracking. This was achieved by assigning and maintaining a static identification number (ID) for each bean as it transited through the camera's field of view. To finalize the class assignment, the system utilized the Majority Voting method, recognizing that an individual bean is captured across multiple frames during its movement. This method determines the final classification based on the highest frequency class predicted within a specific frame window, governed by a "Frames to Confirm Stability" constraint. This constraint dictates that a tracked ID must be continuously detected for a minimum of five frames, and a minimum threshold of three out of those five frames must predict the identical class to validate the final classification output.

Figure 7. Real-time object tracking and class confirmation sequence across four consecutive frames.

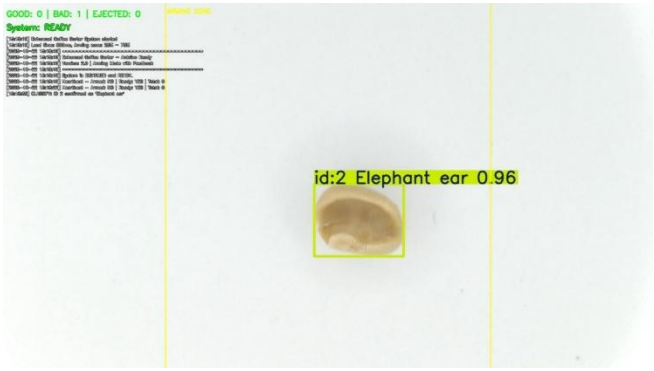


Figure 8. Final class confirmation at the fifth frame using majority voting.

Specifically, the "Frames to Confirm Stability" mechanism initiates its counting sequence at frame one precisely upon the successful detection and assignment of a unique tracking identification number (ID) to an object. The counter increments continuously until the fifth consecutive frame is reached. Only upon satisfying this temporal threshold is the object's classification officially validated. Subsequently, this validated categorical data is transmitted to the downstream control architecture to compute the binary operational decision: either bypassing the mechanical sequence to permit passage or triggering the pneumatic cylinder for ejection.

Figure 9. Example of a confirmed class bounding box transmitted to the pneumatic control unit for actuation.

The establishment of the five-frame threshold is the result of empirical testing conducted to determine the optimal trade-off between decision accuracy and system timing. During the evaluation process, comparative assessments were performed against three-frame and seven-frame thresholds. The empirical results indicated that while a three-frame threshold facilitated rapid system response, the model exhibited susceptibility to visual noise, frequently resulting in erroneous decisions caused by class flickering. Conversely, the implementation of a seven-frame threshold yielded higher prediction consistency but induced accumulated latency. This temporal overhead resulted in an actuation delay, where the pneumatic cylinder deployment lagged behind the actual spatial position of the target bean on the conveyor belt. Consequently, the five-frame threshold was selected as the operational parameter for this prototype, as it provided a measured equilibrium between class confirmation consistency and system response time under the specific experimental conditions of this research.

3. Experimental Results and Discussion

This section presents a systematic experimental evaluation and performance analysis of the automated coffee bean sorting prototype, encompassing the model training phase through to real-world deployment evaluation. Performance evaluation was conducted utilizing standard quantitative metrics, specifically Precision, Recall, F1-score, and mean Average Precision (mAP), alongside an analysis of learning dynamics via loss curve behavior. Furthermore, to establish a comprehensive understanding of multi-class classification behavior, a Confusion Matrix was utilized to evaluate per-class accuracy and specific prediction errors. Ultimately, this research conducted a comparative study of

object detection architectures, namely YOLOv8s, YOLO11s, and YOLOv12s, to identify the model demonstrating the most viable measured balance between algorithmic accuracy and inference speed. Prioritizing empirical operational capability on the physical apparatus is treated as a critical parameter for smart agricultural applications.

To prevent metric ambiguity, this study segregates the evaluation results into three distinct tiers: (1) computational metrics on static datasets (e.g., mAP, precision, recall, and inference time); (2) vision-sensor-only classification performance on the physical moving conveyor; and (3) integrated mechanical rejection performance, which incorporates the physical constraints of the pneumatic hardware.

3.1 The Sim-to-Real Gap: Baseline Performance vs. Real-world Instability

During the initial performance evaluation, the YOLOv8s, YOLO11s, and YOLOv12s architectures were trained on the Baseline dataset, and their convergence behavior was evaluated via loss curves, as depicted in Figure 10. Preliminary testing on the validation set indicated that the YOLOv8s architecture yielded the highest mean Average Precision (mAP@50-95).

Figure 11 presents a trade-off analysis between performance and computational speed for the compared models, with the primary y-axis denoting average inference time per image and the secondary y-axis denoting average mAP@50-95 on the test set. The data illustrates divergent behaviors between the baseline model cohort and the hyperparameter-tuned model. Within the baseline cohort, the YOLOv8s architecture recorded the lowest inference time at 8.95 ms per image. However, the evaluation of the newer architectures in an untuned state (v11s Baseline and v12s Baseline) revealed performance degradation, with mAP decreasing to 0.80 and 0.82, respectively, accompanied by an increase in inference time to 16.48 ms. This outcome indicates that deploying recent architectural iterations without targeted hyperparameter tuning does not inherently yield elevated performance on a highly specific dataset.

Conversely, the hyperparameter-tuned YOLOv12s model demonstrated a significant measured increase in static performance. The data indicates that structural tuning and targeted data augmentation elevated the efficacy of the v12s architecture, increasing the average mAP@50-95 to 0.9553—the highest recorded value compared to the entire baseline cohort. Analyzing the computational latency, the inference time increased to 17.69 ms (a +1.21 ms variance compared to the v12s Baseline), which correlated with a +12.85 percentage point absolute increase in mAP.

This operational point is viable for real-time application on the sorting conveyor; qualitative physical testing indicated that models with higher static-image accuracy generally exhibited more consistent temporal predictions. However, this research has not yet evaluated temporal consistency via direct quantitative metrics under dynamic motion conditions.

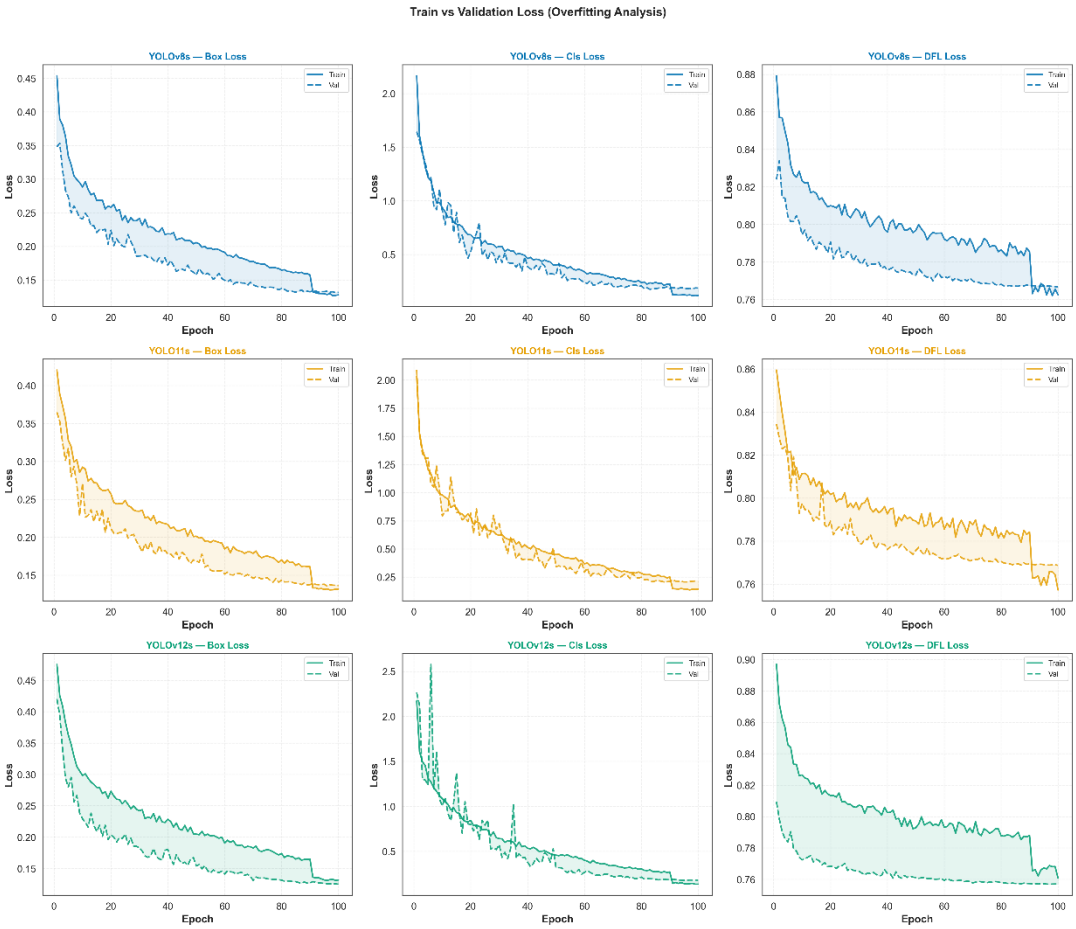


Figure 10. Training and validation loss curves for the baseline models.

The phenomenon where the validation loss remains consistently lower than the training loss throughout the training process does not indicate data leakage or conditions of underfitting or overfitting. Conversely, this phenomenon is an expected convergence behavior within YOLO-family architectures when rigorous regularization mechanisms are applied. This behavior can be elucidated through data engineering and architectural mechanisms, encompassing three primary factors:

1. **Aggressive Data Augmentation Influence:** During the training phase, input image data is subjected to complex spatial and color-space modifications via advanced techniques, specifically Mosaic (0.6), MixUp (0.1), and Random Erasing (0.4). These data distortions generate a challenging learning environment, deliberately elevating the training loss to prevent the model from overfitting to the original data patterns. Conversely, evaluation during the validation phase is conducted on unmodified, original images (clean data). This discrepancy in dataset difficulty enables the model to yield more accurate predictions on the validation set, thereby naturally manifesting a lower validation loss.
2. **Model Tension Balancing Mechanism (Close Mosaic Effect):** The hypothesis regarding the influence of data augmentation was empirically validated via the close_mosaic: 10 hyperparameter configuration, which dictates the cessation of advanced image augmentation during the final 10 training epochs. This mechanism permits the model to fine-tune its weights based on the true data distribution. This phenomenon is distinctly reflected in the training loss curve, which exhibits a sudden drop during the terminal phase to converge with the validation loss trajectory. This provides empirical proof that the preceding loss gap was a direct consequence of the resistance imposed by the regularization mechanisms.

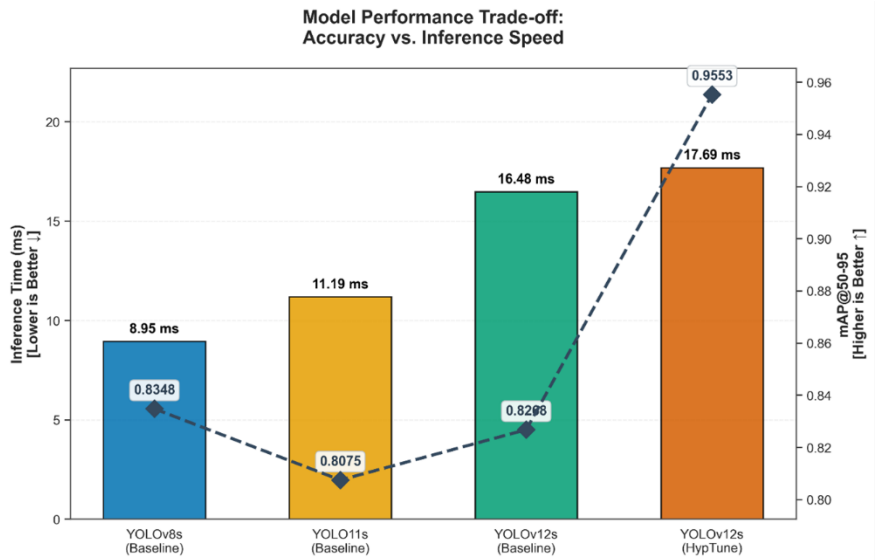


Figure 11. Performance trade-off analysis: mAP@50-95 versus inference time.

To select the core architecture, empirical testing was conducted on the physical moving conveyor belt to evaluate prediction stability, as comparatively illustrated in **Figure 12**. Observational data indicated that the YOLOv8s architecture encountered class label flickering when subjected to object motion blur. Conversely, the YOLOv12s architecture demonstrated continuous temporal consistency by maintaining bounding boxes across video frames. Accepting a marginal reduction in inference speed and static computational accuracy in exchange for the object-tracking stability of YOLOv12s was determined to be an appropriate engineering trade-off. Consequently, the YOLOv12s architecture was selected as the primary structural framework.

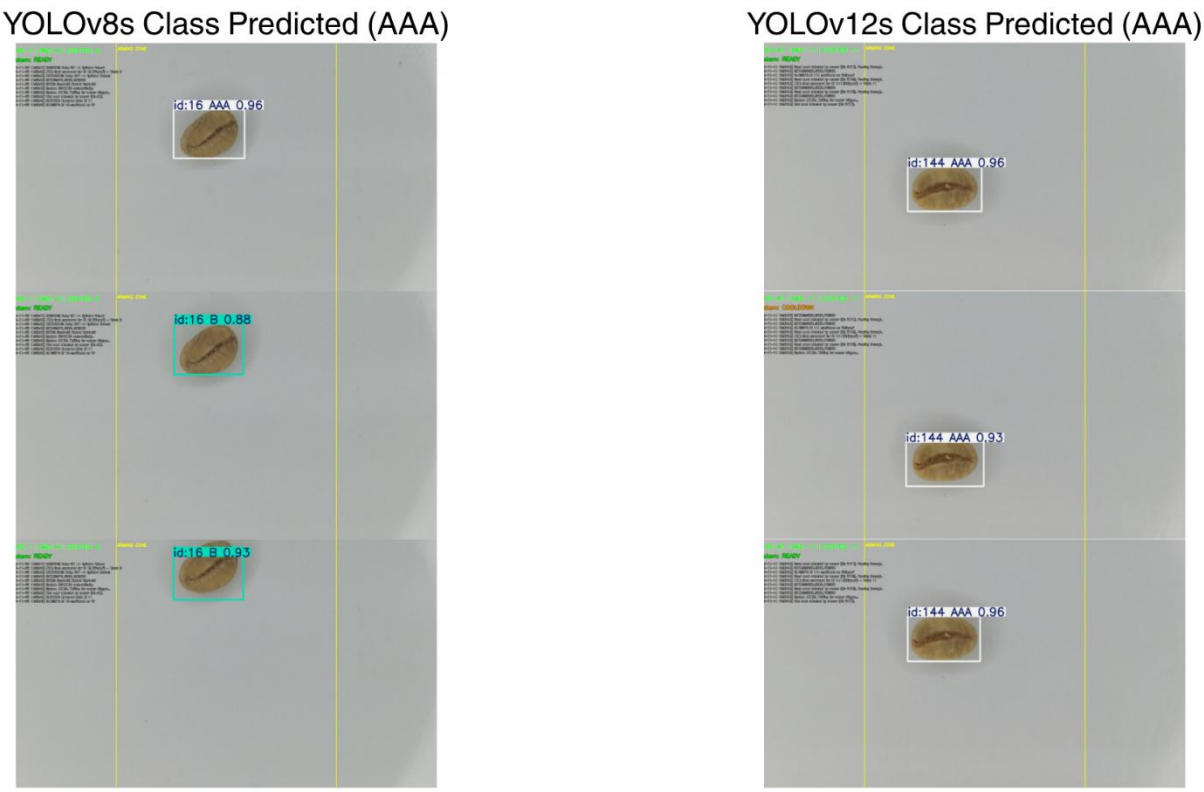


Figure 12. Comparison of class prediction stability between baseline YOLOv8s and YOLOv12s on a dynamic conveyor.

In selecting the neural network architecture for practical deployment, the researchers did not exclusively consider the statistical accuracy metric (mAP) evaluated on static image datasets. Instead, a preliminary real-world testing phase on the physical conveyor belt was incorporated to evaluate prediction stability under dynamic motion conditions. Following the training of the baseline models, the three architectures (YOLOv8s, YOLO11s, and YOLOv12s) were deployed via the vision camera to process and detect moving coffee beans. For this empirical observation, two primary coffee bean classes with distinctly different morphological characteristics were utilized: Class AAA (representing the acceptable bean category) and Class Black (representing the defective bean category). The objective was to observe and compare the decision-making behavior of the models.

Qualitative observations on the active conveyor belt revealed a notable phenomenon. The YOLOv8s architecture, which recorded the highest static mAP, exhibited instability when subjected to the dynamic environment, characterized by bounding box and class label oscillation (class flickering). As comparatively illustrated in Figure 12, this instability can be elucidated through architectural foundations. Specifically, YOLOv8 relies predominantly on convolutional operations constrained by a local receptive field. Consequently, when object motion blur distorts localized spatial features, the network experiences a sudden degradation in prediction confidence.

Conversely, the YOLOv12s architecture is augmented by the integration of an Area Attention Mechanism, which possesses the capacity to evaluate global contextual dependencies rather than relying strictly on localized pixel data. Facilitated by this structural configuration, YOLOv12s demonstrated higher frame-to-frame prediction consistency compared to the baseline counterparts. During qualitative observation under physical conveyor testing, the model exhibited a higher tendency to consistently maintain the bounding box and the initial class decision (Class AAA or Black) across multiple frames of motion. Even when prediction oscillation was observed, the model predominantly grouped the object within the correct macro-category (e.g., a Class AAA prediction oscillating to Class A, AA, or B, all of which remain within the acceptable category).

Frame-to-frame prediction consistency is an essential engineering parameter for this sorting apparatus. The microcontroller (Arduino) requires a consistent data signal to accurately compute the actuation timing for the pneumatic cylinder to eject defective beans. Based on this hardware engineering compatibility and the theoretical architectural advantages, YOLOv12s was selected as the primary network for subsequent advanced hyperparameter tuning.

However, it must be emphasized that this comparative section and the visual data presented in Figure 12 are strictly intended to describe the qualitative behavior of the system under actual dynamic conditions. This research has not evaluated temporal consistency utilizing specific quantitative or statistical metrics. These qualitative observations, which align with underlying architectural theory, were utilized strictly as preliminary evaluation criteria to support engineering decision-making. The development of quantitative metrics for temporal consistency is designated as a challenge for future work.

To resolve the limitations associated with morphologically ambiguous classes, hyperparameter tuning and targeted data augmentation techniques were applied to the YOLOv12s architecture. To evaluate training stability, statistical test results from five random initialization seeds (Seeds: 42, 123, 2023, 3407, 999) were compared between the baseline model and the hyperparameter-tuned model. This evaluation established the mean and standard deviation (SD) for Precision, Recall, mAP50, and mAP50-95 metrics, as summarized in Table 6. The comparative data indicated a significant reduction in standard deviation (variance) across the experimental runs for the hyperparameter-tuned model, denoting an increase in training stability. This correlates with the convergence behavior observed in the loss curves (Figure 13) and the recovery of accuracy in problematic classes, as evidenced by the confusion matrices (Figure 14) before and after tuning.

Table 6. Statistical summary (Mean ± SD) of model performance evaluated on the unseen test set (5 runs)

Models	Precision	Recall	mAP50	mAP50-95
YOLOv8s	0.8118 ± 0.0097	0.7621 ± 0.0235	0.8362 ± 0.0119	0.8348 ± 0.0117
YOLOv11s	0.8113 ± 0.0258	0.7392 ± 0.0221	0.8093 ± 0.0130	0.8075 ± 0.0133
YOLOv12s	0.8110 ± 0.0236	0.7478 ± 0.0114	0.8280 ± 0.0162	0.8268 ± 0.0162
YOLOv12s (Hyperparameter-Tune)	0.9166 ± 0.0082	0.9278 ± 0.0151	0.9616 ± 0.0040	0.9553 ± 0.0040

Table 6 presents the statistical evaluation results on the unseen test set to analyze the generalization ability of the models. The empirical results indicated that the three baseline architectures yielded an mAP@50-95 ranging from 0.80 to 0.83. However, the hyperparameter-tuned YOLOv12s architecture demonstrated an increased mAP@50-95 of 0.9553. Furthermore, the statistical analysis demonstrated that the standard deviation (SD) of the tuned model decreased to 0.0040. This reduction in variance provides empirical evidence indicating that the hyperparameter tuning and data augmentation strategies not only mitigated the misclassification rate but also increased the temporal prediction consistency of the model. This constitutes a critical prerequisite prior to integration with the physical sorting hardware apparatus.

An empirical case study reflecting the impact of the dataset refinement is the classification behavior of the Pea berry class. Analysis of the confusion matrix revealed that in the baseline model, this class exhibited an elevated misclassification rate. This limitation stems from a physical constraint: the spherical morphology of the Pea berry, which differs from the standard flat-sided morphology of typical coffee beans. The utilization of a 2D planar imaging vision system, which captures only a single projection, results in the loss of volumetric data necessary for accurate morphological differentiation.

Upon implementing a data-centric mitigation approach—specifically, incorporating supplementary images of the problematic class from multiple angular perspectives into the hyperparameter-tuned dataset—the model acquired the capacity to extract the latent spatial features. Consequently, the classification accuracy for the Pea berry class recovered and increased, as evidenced by the post-tuning confusion matrix results presented in **Figure 14.**

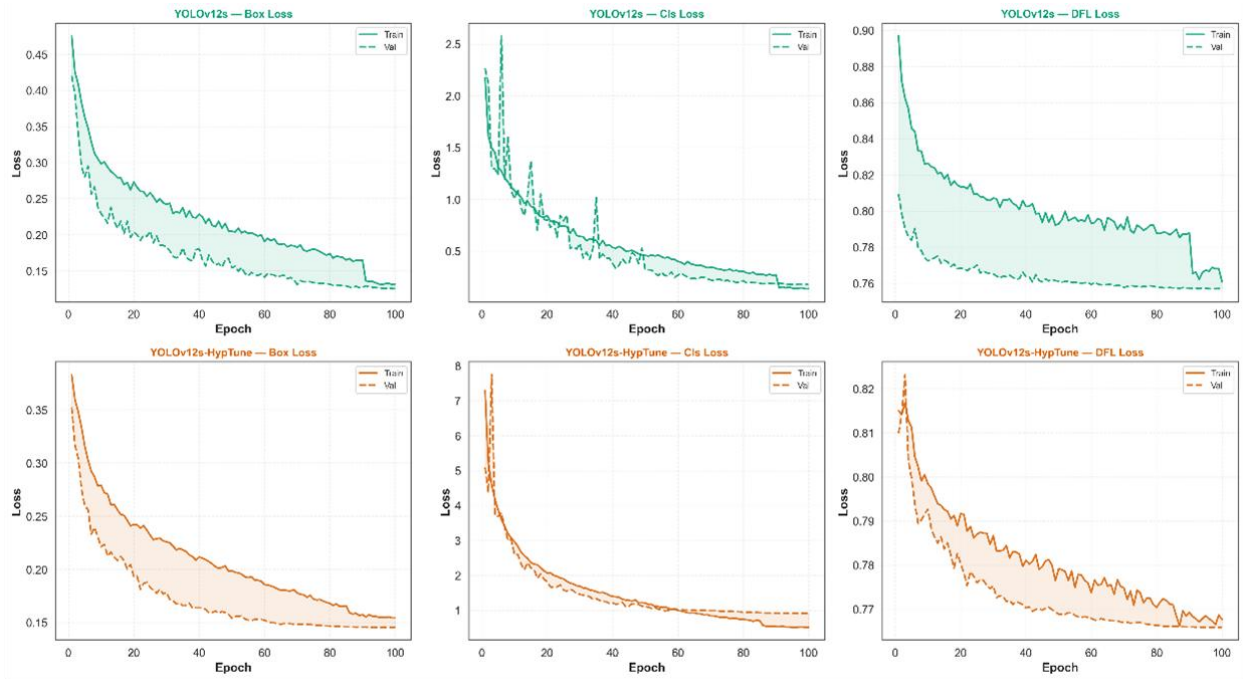
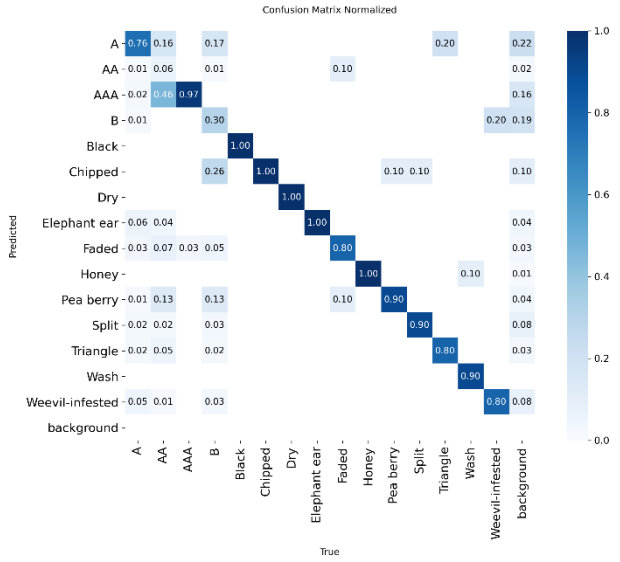


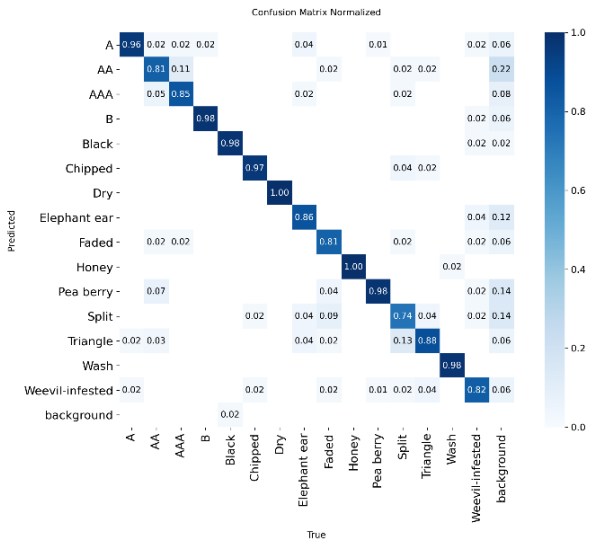
Figure 13. Training and validation loss curves: Baseline versus hyperparameter-tuned YOLOv12s.

YOLOv12s (Baseline)



(a)

YOLOv12s (Hyperparameter-Tune)



(b)

Figure 14. Confusion matrices evaluated on the unseen test set: YOLOv12s Baseline (a) and YOLOv12s Hyperparameter-Tuned (b).

A notable limitation was observed in the detection of the Pea berry class. This limitation primarily stems from the physical morphology of this specific class, which exhibits a spherical shape, contrasting with the flat-sided morphology of standard coffee beans. Compounded by the top-down 2D perspective of the vision camera installation, the model lacks the volumetric depth data and spatial features requisite for accurate morphological feature extraction. This constraint resulted in an elevated False Negative rate, as illustrated in the baseline evaluation of **Figure 14**.

3.2 Ablation Study on Advanced Training Strategies

To analyze the significance of the training strategies, an ablation study was conducted to compare the vanilla baseline YOLOv12s architecture with the hyperparameter-tuned model. This study isolates the empirical effects of two primary components as follows:

1. Impact of Structural Hyperparameter Tuning: Utilizing an input resolution of 640x640 pixels in the baseline model induced a feature extraction bottleneck for minute defects, resulting in the mAP@50-95 stagnating at 0.82. Scaling the input resolution to 704x704 pixels increased small-object spatial awareness and elevated the measured mAP@50-95 to 0.9553. This validates the necessity of aligning parameter boundaries with the physical dimensions of the target objects.

2. Impact of Targeted Data Augmentation: When evaluated in the actual operational environment, the baseline model—which lacked simulated variance during training—exhibited susceptibility to conveyor dynamics. This resulted in 178 false rejects out of a standardized physical test batch of 1,500 coffee beans (comprising 100 beans for each of the 15 evaluation classes). The integration of advanced augmentation techniques increased the model's tolerance to physical noise, resulting in a reduction of false rejects to 96 within the identical sample size (a 46% decrease). Concurrently, the vision-only sorting accuracy increased from 83.07% to 89.60%.

Summary: Empirical data from this study indicates that mitigating the sim-to-real gap cannot rely exclusively on the core algorithmic architecture. It necessitates a synergistic effect between structural parameter scaling for deep feature extraction and the application of augmentation techniques to establish tolerance against dynamic environments.

3.3 Real-World Sorting Performance (Vision Only)

The performance evaluation of the sorting apparatus in the operational environment is divided into two primary phases: (1) vision-sensor-only evaluation, and (2) integrated mechanical rejection performance, the latter of which is discussed in the subsequent section.

During the initial evaluation phase (vision-sensor-only evaluation), the conveyor belt test results for the baseline and hyperparameter-tuned models distinctly reflected the sim-to-real gap, as illustrated by the confusion matrices in **Figure 14.**

It is imperative to note that the performance improvement observed in this phase is the result of a combined hardware-software resolution scaling effect. During the baseline model testing phase, the vision system processed images via camera hardware set to a 720p resolution. To physically align with the algorithmic input resolution scaling (from 640 to 704 pixels) and to prevent a physical hardware bottleneck in detecting minute defects, the camera configuration was upgraded to a 1080p resolution during the hyperparameter-tuned model evaluation phase. Consequently, the reported increase in real-world vision sorting accuracy—from 83.07% (Baseline) to 89.60% (Hyperparameter-Tuned)—must be interpreted as the synergistic outcome of optimizing both the computational hyperparameter boundaries and the physical sensor resolution, rather than an isolated algorithmic improvement (detailed in **Tables 7, 9, 10, and Figure 15**).

However, the 89.60% accuracy achieved by the vision system cannot be directly translated into net mechanical rejection accuracy. The theoretical maximum capacity of the algorithmic model is attenuated by the physical limitations of the pneumatic hardware, specifically the actuation latency of the pneumatic valves and the spatial variance of the objects during transit. These physical factors constitute the primary bottleneck of the apparatus.

Table 7. Vision-sensor only real-world classification performance using the baseline model.

Class	Model Predicted		Testing Duration
	Good	Defect	(mm:ss)
A (Good)	66	34	05:14
AA (Good)	82	18	04:54
AAA (Good)	51	49	04:20
B (Good)	88	12	04:30
Dry (Good)	97	3	04:59
Honey (Good)	100	0	04:49
Pea berry (Good)	42	58	04:57
Wash (Good)	96	4	04:40
Black (Defect)	4	96	04:34
Chipped (Defect)	14	86	04:41
Elephant ear (Defect)	6	94	05:08
Faded (Defect)	16	84	04:52
Split (Defect)	14	86	04:31
Triangle (Defect)	15	85	04:41
Weevil-infested (Defect)	7	93	04:45
Defect Overall Accuracy			89.14%
Good Overall Accuracy			77.75%

Total Accuracy	83.07%

As detailed in Table 7, the baseline model demonstrated an overall vision-based sorting accuracy of 83.07%. During the dynamic evaluation, constrained by the manual feeding mechanism of the current prototype, the vision system recorded an average testing duration of approximately **286 seconds per 100-bean class**, translating to an effective processing throughput of approximately **0.35 beans per second**. Furthermore, the classification accuracy for acceptable (Good) beans was measured at 77.75%, whereas the accuracy for defective beans was **89.14%**. These observed behaviors indicate that while the baseline model yielded satisfactory outcomes in detecting defective beans, misclassifications persisted in differentiating morphologically similar acceptable categories. Notably, the AAA and Pea berry classes exhibited comparatively lower accuracy, highlighting the limitations of the baseline architecture when processing spatially ambiguous features on a moving conveyor.

Based on the physical operational testing of the baseline model, limitations were observed in classifying classes with ambiguous physical morphologies. To address these limitations and elevate operational capacity, **a systematic hardware-software optimization strategy** was executed according to the following methodologies.

1. **Multi-angle View Expansion:** Supplementary images were acquired specifically for the problematic classes (A, AA, AAA, B, Pea berry, and Chipped). Each target bean was recorded across four physical orientations (standard projection, 90-degree rotation, and inverted) to augment morphological variance, adding 200 images per class.

2. **Targeted Data Augmentation:** Data augmentation techniques were applied via the Roboflow platform exclusively to the training set to increase model tolerance against dynamic operational conditions.

3. **Structural Hyperparameter Optimization:** The input image resolution was scaled to preserve minute object details. The learning rate and the intensity parameters of the augmentation scripts were adjusted to synchronize with the modified dataset.

4. **Hardware Upgrade:** The vision sensor hardware was upgraded from a 720p to a 1080p recording resolution to facilitate micro-feature extraction on the moving conveyor belt.

Table 8. Comparison of image quantities for problematic classes before and after targeted data augmentation

Class	Baseline	Hyperparameter-Tune
A (Good)	305	505
AA (Good)	300	500
AAA (Good)	301	501
B (Good)	300	500
Pea berry (Good)	305	505
Chipped (Defect)	304	504

Table 9. Vision-sensor only real-world classification performance using the hyperparameter-Tuned model.

Class	Model Predicted		Testing Duration
	Good	Defect	(mm:ss)
A (Good)	75	25	04:58
AA (Good)	94	6	04:54
AAA (Good)	87	13	04:58
B (Good)	73	27	03:54
Dry (Good)	100	0	05:10
Honey (Good)	100	0	04:09
Pea berry (Good)	87	13	04:30
Wash (Good)	88	12	04:40
Black (Defect)	0	100	04:54
Chipped (Defect)	3	97	04:49
Elephant ear (Defect)	2	98	04:38
Faded (Defect)	26	74	04:54
Split (Defect)	2	98	05:18
Triangle (Defect)	16	84	04:38
Weevil-infested (Defect)	11	89	04:49
Defect Overall Accuracy			91.43%
Good Overall Accuracy			88%
Total Accuracy			89.6%

The application of hyperparameter tuning and targeted data augmentation to the baseline model aimed to mitigate limitations in classifying classes exhibiting morphological ambiguity. The performance evaluation on the dynamic sorting hardware, as summarized in Table 9, indicated a measurable increase in accuracy. The hyperparameter-tuned model demonstrated an average sorting accuracy of 88.00% for the acceptable bean category and 91.43% for the defective category. These metrics represent a concrete improvement over the baseline model, particularly within the premium-grade coffee bean categories (Classes A, AA, and AAA), which exhibited an increased correct classification rate across all respective classes.

Based on the comparative analysis of prediction behavior in Figure 15, a primary engineering finding is that the tuned model mitigated the loss of acceptable yield by reducing the false reject rate by 46% (decreasing from 178 to 96 beans out of the 1,500-bean physical test batch). Concurrently, the model achieved a reduction in the false accept rate of defective beans. This empirical evidence indicates that aligning the model optimization strategy with the dynamics of the physical operational environment not only elevated statistical accuracy but also served as a critical factor in yield preservation. This aligns with the operational and economic constraints requisite for artificial intelligence applications within physical production environments.

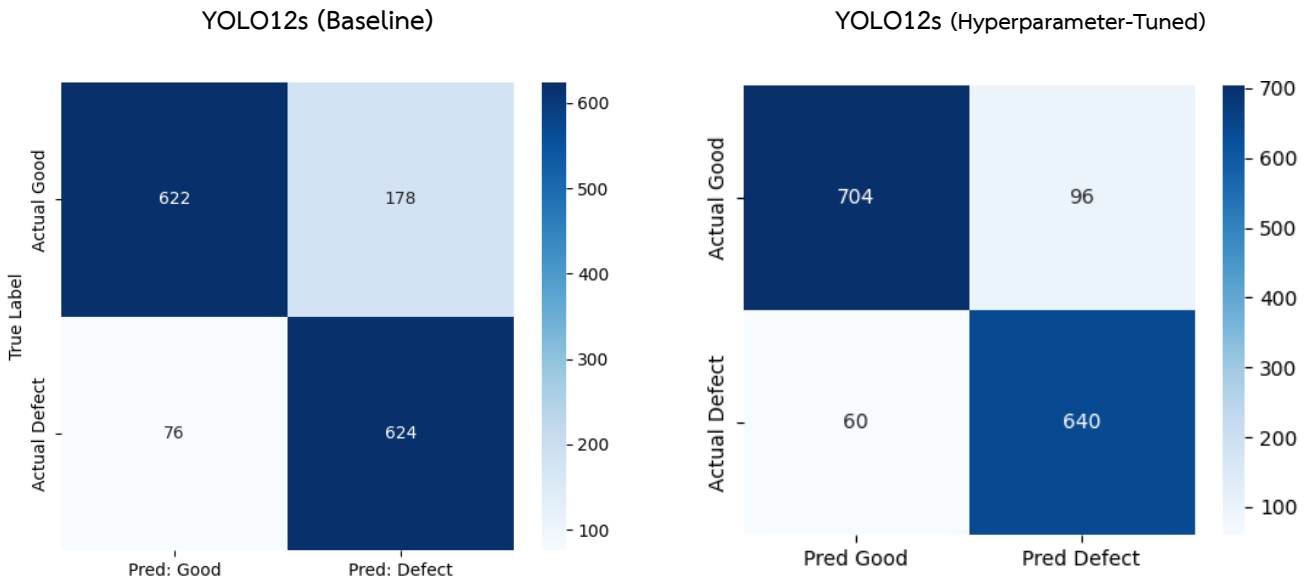


Figure 15. Comparative analysis of real-world prediction outcomes (Vision-sensor only), highlighting the reduction of False Rejects (overkill) and False Accepts after hyperparameter tuning.

It must be emphasized that the data reported in **Table 10** strictly represents a vision-sensor-only evaluation conducted on the physical moving conveyor belt. These results do not incorporate physical perturbation variables, specifically pneumatic valve actuation laten cy, the spatial displacement of objects in transit, and the mechanical constraints of the pneumatic ejection apparatus. The evaluation of these hardware-induced impacts is isolated and will be presented separately within the integrated system evaluation in the subsequent section.

Table 10. Vision-sensor only Real-World Sorting Result Baseline vs Hyperparameter -Tune

Class	Type	Baseline (%)	Hyperparameter -Tune (%)	Δ(%)	Result
A	Good	66	75	+9	up
AA	Good	82	94	+12	up
AAA	Good	51	87	+36	up
B	Good	88	73	-15	drop
Dry	Good	97	100	+3	up
Honey	Good	100	100	+0	same
Pea berry	Good	42	87	+45	up
Wash	Good	96	88	-8	drop
Black	Defect	96	100	+4	up
Chipped	Defect	86	97	+11	up
Elephant ear	Defect	94	98	+4	up
Faded	Defect	84	74	-10	drop
Split	Defect	86	98	+12	up
Triangle	Defect	85	84	-1	drop
Weevil-infested	Defect	93	89	-4	drop
Good Overall	-	77.8	88.0	+10.2	up
Defect Overall	-	89.14	91.4	+2.3	up
Total Accuracy	-	83.07	89.6	+6.5	up

To systematically analyze the sim-to-real gap, a comparative evaluation of detection performance was conducted, as illustrated in **Figure 16**. This figure presents a comparison of the evaluation metrics obtained from the static dataset (unseen software benchmark) against the operational results acquired during dynamic physical deployment integrated with the pneumatic hardware.

The empirical results indicated a distinct sim-to-real gap, evidenced by the performance degradation of the baseline model. This phenomenon is attributed to dynamic variables within the physical environment, specifically incident lighting variance, bounding box flickering during conveyor transit, and hardware latency. These variables constitute operational constraints that are absent from the static simulated dataset environment.

Conversely, the data presented in Figure 16 indicates that the optimized YOLOv12s model, following structural hyperparameter tuning and targeted data augmentation, maintained a real-world vision-sensor prediction accuracy approaching 90% (specifically 89.60%). This operational level approximates the initial software-based evaluation results. This empirical evidence indicates that the proposed optimization strategy mitigates the effects of dynamic image instability and reduces the sim-to-real gap. Consequently, this enables the vision system to transmit actuation signals to the pneumatic mechanism with increased temporal consistency under physical constraints.

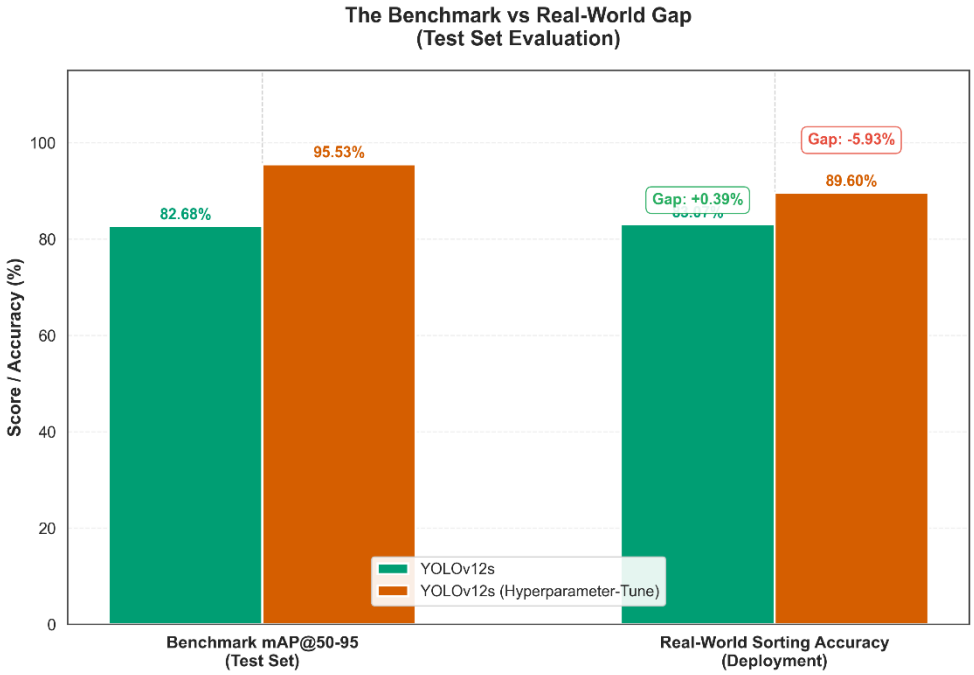


Figure 16. Performance comparison: Software benchmark (Test Set) versus real-world vision sorting accuracy.

The sim-to-real gap phenomenon and the outcomes of the optimization strategy are comparatively illustrated in Figure 16. Empirical data reveals a crucial engineering trade-off. while deployment into the physical environment introduced a sim-to-real performance gap of approximately **5.93%** (measured as a decrease from the static mAP@50-95 of 95.53% down to a real-world vision accuracy of 89.60%), the hyperparameter-tuned model demonstrated significantly higher temporal stability. Ultimately, this optimized state mitigated cross-frame flickering and provided the necessary tolerance to the dynamic operational environment required for successful pneumatic integration, as detailed in **Table 10**. The dynamic hardware evaluation presented in Table 9 confirms that the hyperparameter-tuned model yielded an average classification accuracy of 88.00% for the acceptable bean category and 91.43% for the defective category. This represents a measurable comparative improvement over the baseline model, particularly within the premium-grade categories (Classes A, AA, and AAA), which demonstrated increased correct classification rates across all respective classes.

A primary engineering finding is that the optimized model reduced the false reject rate by 46%, constituting a critical factor for yield preservation within industrial sorting processes. Benchmarking the dynamic vision sorting accuracy of 89.60% against relevant literature highlights a critical engineering perspective regarding metric interpretation. Although previous studies, such as those by Chang and Huang [1] or Chen et al. [3], reported accuracy levels ranging from 94.0% to 98.0%, those figures

were derived from static image evaluation under controlled illumination conditions, or relied on high-cost sensor technologies, such as hyperspectral imaging, without integrating dynamic visual challenge variables into the evaluation framework.

In contrast, the 89.60% dynamic vision accuracy achieved in this research is the outcome of a conveyor-based evaluation that incorporates visual perturbations, specifically motion blur and the requirement for frame-to-frame temporal consistency. This empirical evidence indicates that integrating the YOLOv12s architecture with the proposed optimization strategy mitigates the visual sim-to-real gap under the constraints of a standard RGB vision infrastructure.

3.4 End-to-End System Evaluation (Pneumatic Actuation Integrated)

During the final evaluation phase, the vision system was integrated with the pneumatic actuator mechanism to assess the comprehensive operational performance of the prototype. The mechanical rejection performance test results, detailed in Table 11, indicated that the apparatus achieved an overall net mechanical rejection accuracy of 73.24%. This significant degradation in accuracy, relative to the vision system (89.60%), is a direct consequence of physical hardware constraints, primarily the actuation latency of the pneumatic mechanism and the spatial displacement of objects in transit.

To quantify the performance degradation induced by hardware integration, comparative data between the vision-sensor-only evaluation (Trial 1) and the fully integrated pneumatic evaluation (Trial 2) of the hyperparameter-tuned model was analyzed. The system exhibited a measured variance in operational performance, with the sorting accuracy decreasing from 89.60% in Trial 1 to a net mechanical rejection accuracy of 76.0% in Trial 2. These empirical results explicitly indicate that the operational consistency of the pneumatic hardware remains a critical bottleneck and a primary limitation of the prototype, necessitating future mechanical refinement.

Table 11. End-to-end mechanical rejection performance combining the hyperparameter-Tuned model and pneumatic actuation.

Class	Model Predicted		Actual Rejected (Machine)
	Good	Defect	
A (Good)	89	11	11
AA (Good)	93	7	7
AAA (Good)	89	11	8
B (Good)	77	23	22
Dry (Good)	99	1	1
Honey (Good)	100	0	0
Pea berry (Good)	86	14	13
Wash (Good)	89	11	10
Black (Defect)	18	82	57
Chipped (Defect)	35	66	54
Elephant ear (Defect)	31	69	37
Faded (Defect)	61	39	30
Split (Defect)	13	87	42
Triangle (Defect)	65	35	34

Weevil-infested (Defect)	59	41	38
Defect Overall Accuracy			59.86%
Good Overall Accuracy			90.25%
Total Accuracy			76.07%
Mechanical Actuation Success Rate			73.24%

The results of the second trial revealed that the defect overall accuracy decreased to 59.86%. This performance degradation phenomenon serves as empirical evidence reflecting the sensitivity of the vision system to physical environment drifts during continuous operation. Variables such as illumination variations, fluctuations in conveyor velocity, spatial distribution variance among the beans, and the limitation of the sensor resolution in detecting micro-sized beans reside outside the control parameters of the software architecture.

Furthermore, the reduction in the overall net mechanical rejection accuracy to 73.24% highlights a separate mechanical limitation: accumulated pressure drop within the pneumatic compressor system during continuous actuation sequences. These findings emphasize that mitigating the sim-to-real gap to an industrially acceptable level cannot rely exclusively on artificial intelligence algorithm optimization. It concomitantly requires the integration of environmental control systems and the implementation of highly stable, industrial-grade mechanical mechanisms.

As detailed in Table 12, the automated conveyor-based coffee bean sorting prototype utilizing deep learning achieved a measured average throughput of 236.8 grams per hour. Accuracy limitations persist in specific edge cases, notably the classification of morphologically analogous beans or when image acquisition is subjected to uneven illumination and shadowing, which can induce classification errors within the neural network. This limitation aligns with existing image processing literature, which establishes that dataset quality and the physical data acquisition environment directly influence the operational performance of the model.

Table 12. Processing capacity (grams/hour) of the end-to-end automated sorting system.

Class	Throughput (g/h)
A (Good)	228
AA (Good)	216
AAA (Good)	288
B (Good)	168
Dry (Good)	336
Honey (Good)	276
Pea berry (Good)	192
Wash (Good)	252
Black (Defect)	252
Chipped (Defect)	204
Elephant ear (Defect)	204
Faded (Defect)	216
Split (Defect)	204
Triangle (Defect)	276
Weevil-infested (Defect)	240
Average	236.8

Regarding the system throughput, continuous sorting evaluations demonstrated an average capacity of 236.8 grams per hour. Although the proposed prototype successfully validated integrated operational capabilities and tangibly reduced the sim-to-real gap, the current processing

rate reflects a laboratory-scale constraint. This constraint is primarily governed by the manual feeding process rather than the theoretical limitations of the deployed artificial intelligence architecture.

The hyperparameter-tuned YOLOv12s model achieved an inference time of 17.69 milliseconds per frame (equivalent to approximately 56 frames per second). If an automated vibratory feeder were integrated to maintain optimal spatial separation between the coffee beans, the vision system possesses sufficient computational capacity to process 20 to 30 beans per second per lane. Consequently, the theoretical throughput of a single-lane system could linearly scale to approximately 10 to 15 kilograms per hour. This effectively shifts the system bottleneck from software computational capacity to mechanical handling limits.

However, as the feed rate scales to industrial levels, the true system bottleneck will transition from visual processing to mechanical actuation recovery latency. The pneumatic cylinder utilized in the prototype exhibits inherent physical latency, comprising the required duration for the extension and retraction of the piston rod. If defective beans transit sequentially within a timeframe shorter than this mechanical recovery period, the mechanism cannot respond adequately, resulting in defective bean leakage (false accepts). Therefore, to achieve industrial-grade throughput (e.g., metric tons per hour), scaling the system necessitates transitioning from physical pneumatic cylinders to arrays of high-frequency air-jet nozzles integrated with a wide-chute (free-fall) mechanism.

Furthermore, migrating the processing unit from a standard personal computer to a specialized Edge AI hardware platform (e.g., NVIDIA Jetson) is imperative to mitigate the effects of non-real-time operating system (OS) jitter. Such architectural modifications will support the YOLOv12s model in maintaining highly stable processing speeds, which is a critical prerequisite for practical industrial deployment.

5. Conclusion

5.1. Summary of Findings

This research presents an experimentally validated automated coffee bean sorting prototype capable of classifying 15 distinct morphological categories. The empirical results tangibly demonstrate the challenges inherent in the sim-to-real gap. While the baseline model exhibited satisfactory static image accuracy, it manifested temporal instability when subjected to the dynamic environment of the moving conveyor, resulting in false mechanical rejections, particularly among morphologically analogous bean categories.

The experimental findings indicate that the implementation of the YOLOv12s architecture, synergized with aggressive hyperparameter tuning, observably mitigated these dynamic instabilities. Empirical data revealed a crucial engineering trade-off. bridging the deployment environment introduced a sim-to-real performance gap of approximately 5.93% (measured from a static mAP@50-95 of 95.53% down to a real-world vision accuracy of 89.60%). However, this optimized state demonstrated significantly higher temporal stability and mitigated cross-frame flickering on the physical conveyor, making it superior for pneumatic integration. When evaluated alongside the integrated pneumatic mechanism, operational performance remained constrained by physical hardware limitations. The system recorded a net vision sorting accuracy of 76.07% during integrated operation, with the overall mechanical actuation success rate restricted to 73.24%.

5.2. Generalizable Insights and System Dynamics

Based on the integrated prototype evaluation, this research establishes generalizable engineering insights applicable to analogous automated agricultural sorting systems:

- 5.2.1. Impact of Temporal Instability on Control Error: Hardware actuation inaccuracies are directly correlated with class flickering, which disrupts the precise timing required for pneumatic valve actuation. The application of the Area Attention mechanism inherent in the YOLOv12 architecture contributes to mitigating cross-frame flickering, thereby functioning as a software-level intervention to reduce downstream mechanical actuation errors.

- 5.2.2. Mechanisms of Augmentation in Sim-to-Real Gap Reduction: Simulating motion blur and employing MixUp augmentation forces the neural network to adapt to the spatial and temporal imperfections genuinely encountered on a moving conveyor. Consequently, the network exhibits observed robust behavior against incident illumination fluctuations and dynamic physical perturbations.

- 5.2.3. Correlation Between Resolution Scaling and Micro-Defect Recovery: Scaling the input resolution to 704 pixels emerged as the operational sweet spot. The data explicitly indicates that minute surface defects ignored by the 640-pixel baseline model were successfully recovered at the 704-pixel resolution, while the corresponding inference latency remained compliant with the stringent temporal constraints of the pneumatic control loop.

5.3. Limitations and Future Works

Although the proposed prototype empirically validates the engineering feasibility of integrating deep learning software with mechanical sorting hardware, the evaluation identified specific physical limitations. The net discrepancy between the artificial intelligence classification and the actual mechanical rejection (reflected by the 73.24% mechanical rejection accuracy) stems primarily from the spatial detection limits of the photoelectric sensors regarding micro-sized beans and the inherent physical actuation delays of the pneumatic cylinders. Furthermore, the prototype's current processing throughput of 236.8 grams per hour remains a mechanical bottleneck induced strictly by manual feeding protocols.

To elevate this architectural framework into a fully comprehensive industrial solution, future research trajectories must prioritize structural refinement across both mechanical and computational hardware dimensions. Mechanically, the integration of an automated vibratory feeder is imperative to maintain uniform spatial separation between transiting objects.

Computationally, the prototype currently relies on laptop-based processing combined with standard vision camera hardware. This configuration inherently introduces data transmission latency and non-real-time operating system (OS) jitter. Consequently, subsequent developmental phases must transition toward dedicated Edge AI computing platforms coupled with high-speed industrial machine vision cameras. Upgrading the hardware infrastructure will mitigate data transmission bottlenecks, significantly reduce end-to-end system latency, and efficiently scale the processing capacity to accommodate industrial-grade agricultural yield volumes.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

The dataset and source code used for object tracking, and pneumatic hardware integration have been made publicly available to support reproducibility. The repository can be accessed at: https://github.com/salmonsalmonboi/Coffee-Bean-Sorting-System

CRediT authorship contribution statement

Supakorn Sirimueangmoon: Methodology, Software, Data curation, Investigation, Writing – original draft. Chetsada Jinamoon: Resources, Investigation, Visualization. Audsadakorn Jeerahut: Investigation, Data curation. Suphaloek Khamkoon: Data curation. Pranote Pookkapund: Supervision, Validation. Kwanchai Eurviriyanukul: Supervision, Project administration, Writing – review & editing.

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